

HW 3

Townsend 3.2 (5 pts)

Using the $|+z\rangle$ and $|-z\rangle$ states of a spin- $\frac{1}{2}$ particle as a basis, set up and solve as a problem in matrix mechanics the eigenvalue problem for $\hat{S}_n = \hat{\mathbf{S}} \cdot \mathbf{n}$, where the spin operator $\hat{\mathbf{S}} = \hat{S}_x \mathbf{i} + \hat{S}_y \mathbf{j} + \hat{S}_z \mathbf{k}$ and $\mathbf{n} = \sin \theta \cos \phi \mathbf{i} + \sin \theta \sin \phi \mathbf{j} + \cos \theta \mathbf{k}$. Show that the eigenstates may be written as

$$|+\mathbf{n}\rangle = \cos \frac{\theta}{2} |+\mathbf{z}\rangle + e^{i\phi} \sin \frac{\theta}{2} |-\mathbf{z}\rangle$$

$$|-\mathbf{n}\rangle = \sin \frac{\theta}{2} |+\mathbf{z}\rangle - e^{i\phi} \cos \frac{\theta}{2} |-\mathbf{z}\rangle$$

Rather than simply verifying that these are eigenstates by substituting into the eigenvalue equation, obtain these states by directly solving the eigenvalue problem, as in Section 3.6.

In the spin J_z spin $\frac{1}{2}$ basis, we can find the representation of $\hat{J}_+$ and $\hat{J}_-$, then use the fact that

$$\hat{S}_x = \frac{\hat{J}_+ + \hat{J}_-}{2} \quad (1)$$

$$\hat{S}_y = \frac{\hat{J}_+ - \hat{J}_-}{2i} \quad (2)$$

$$(3)$$

$$S_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (4)$$

The basis states are

$$|\frac{1}{2}, -\frac{1}{2}\rangle \text{ which I write as } \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

and

$$|\frac{1}{2}, \frac{1}{2}\rangle \text{ which I write as } \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\hat{J}_+ |\frac{1}{2}, \frac{1}{2}\rangle = 0 \quad (5)$$

$$\hat{J}_+ |\frac{1}{2}, -\frac{1}{2}\rangle = \hbar \sqrt{j(j+1) - m(m+1)} |\frac{1}{2}, \frac{1}{2}\rangle \quad (6)$$

$$= \hbar \sqrt{\frac{1}{2} \frac{3}{2} + \frac{1}{2} \left(\frac{1}{2}\right)} |\frac{1}{2}, \frac{1}{2}\rangle \quad (7)$$

$$= \hbar \quad (8)$$

We can now find the matrix for $\hat{J}_+$ as

$$\hat{J}_+ = \hbar \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad (9)$$

This gives $\hat{J}_-$ as

$$\hat{J}_- = \hbar \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \quad (10)$$

$$\hat{S}_x = \frac{\hat{J}_+ + \hat{J}_-}{2} \quad (11)$$

$$= \frac{\hbar}{2} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad (12)$$

$$\hat{S}_y = \frac{\hat{J}_+ - \hat{J}_-}{2i} \quad (13)$$

$$= \frac{\hbar}{2} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad (14)$$

The three spin matrices are mutually orthogonal, so we can take the dot product by simply separating out and adding terms

$$\hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (15)$$

$$\hat{S}_x = \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (16)$$

$$\hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (17)$$

$$\mathbf{n} = \sin \theta \cos \phi \mathbf{i} + \sin \theta \sin \phi \mathbf{j} + \cos \theta \mathbf{k}.$$

$$\hat{S}_x \hat{j} \cdot n = \sin \theta \cos \phi \frac{\hbar}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (18)$$

$$\hat{S}_y \hat{j} \cdot n = \sin \theta \sin \phi \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (19)$$

$$\hat{S}_z \hat{k} \cdot n = \frac{\hbar}{2} \cos \theta \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (20)$$

Eigenstates of $\hat{S}$ satisfy

$$\hat{S} \vec{n} = s \vec{n} \quad (21)$$

So the eigenvalue problem is

$$\frac{\hbar}{2} \left[\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \sin \theta \cos \phi + \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \sin \theta \sin \phi + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cos \theta \right] \begin{pmatrix} \langle +z | +n \rangle \\ \langle -z | +n \rangle \end{pmatrix} = \frac{\hbar}{2} n \begin{pmatrix} \langle +z | +n \rangle \\ \langle -z | +n \rangle \end{pmatrix} \quad (22)$$

We can simplify this monstrous equation

$$\begin{pmatrix} \cos \theta & \sin \theta \cos \phi - i \sin \theta \sin \phi \\ \sin \theta \cos \phi + i \sin \theta \sin \phi & -\cos \theta \end{pmatrix} \begin{pmatrix} \langle +z | +n \rangle \\ \langle -z | +n \rangle \end{pmatrix} = n \begin{pmatrix} \langle +z | +n \rangle \\ \langle -z | +n \rangle \end{pmatrix} \quad (23)$$

$$\begin{pmatrix} \cos \theta - n & \sin \theta \cos \phi - i \sin \theta \sin \phi \\ \sin \theta \cos \phi + i \sin \theta \sin \phi & -\cos \theta - n \end{pmatrix} \begin{pmatrix} \langle +z | +n \rangle \\ \langle -z | +n \rangle \end{pmatrix} = 0 \quad (24)$$

$$\begin{pmatrix} \cos \theta - n & \sin \theta (\cos \phi - i \sin \phi) \\ \sin \theta (\cos \phi + i \sin \phi) & -\cos \theta - n \end{pmatrix} = 0 \quad (25)$$

$$\begin{pmatrix} \cos \theta - n & \sin \phi e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta - n \end{pmatrix} = 0 \quad (26)$$

This can be solved by setting the determinant to zero.

We have

$$(\cos \theta - n)(-\cos \theta - n) - \sin^2 \theta = 0 \quad (27)$$

$$-\cos^2 \theta - \cos \theta n + \cos \theta n + n^2 - \sin^2 \theta = 0 \quad (28)$$

$$-\cos^2 \theta + n^2 - \sin^2 \theta = 0 \quad (29)$$

$$-(\cos^2 \theta + \sin^2 \theta) + n^2 = 0 \quad (30)$$

$$n^2 = 1 \quad (31)$$

$$n = \pm 1 \quad (32)$$

We can plug this eigenvalue to find the eigenstates.

(33)

$$\begin{pmatrix} \cos \theta - n & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta - n \end{pmatrix} = 0 \quad (34)$$

(35)

0.0.1 $n = 1$

$$\begin{pmatrix} \cos \theta - 1 & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta - 1 \end{pmatrix} \begin{pmatrix} \langle +z | +n \rangle \\ \langle -z | +n \rangle \end{pmatrix} = \vec{0} \quad (36)$$

(37)

We have two equations from this

$$(\cos \theta - 1) \langle +z | +n \rangle + \sin \theta e^{-i\phi} \langle -z | +n \rangle = 0 \quad (38)$$

$$\sin \theta e^{i\phi} \langle +z | +n \rangle - (\cos \theta + 1) \langle -z | +n \rangle = 0 \quad (39)$$

These two equations provide the same information, I'll use the first arbitrarily. We can use the half angle identities to simplify.

$$\sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1 - \cos \theta}{2}} \quad (40)$$

$$\rightarrow 1 - \cos \theta = 2 \sin^2 \frac{\theta}{2} \quad (41)$$

$$\rightarrow \cos \theta - 1 = -2 \sin^2 \frac{\theta}{2} \quad (42)$$

This becomes

$$-2 \sin^2 \frac{\theta}{2} \langle +z | +n \rangle = -\sin \theta e^{i\phi} \langle -z | +n \rangle \quad (43)$$

We also have the trig identity (RAAGH THIS TOOK SO LONG TO REALIZE AAAA, WHOOPS)

$$\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \quad (44)$$

$$-2 \sin^2 \frac{\theta}{2} \langle +z | +n \rangle = -2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{i\phi} \langle -z | +n \rangle \quad (45)$$

$$\langle +z | +n \rangle = \frac{\cos \frac{\theta}{2}}{\sin \frac{\theta}{2}} e^{i\phi} \langle -z | +n \rangle \quad (46)$$

$$\tan \frac{\theta}{2} e^{-i\phi} \langle +z | +n \rangle = \langle -z | +n \rangle \quad (47)$$

We have to satisfy the condition that

$$|\langle +z | +n \rangle|^2 + |\langle -z | +n \rangle|^2 = 1 \quad (48)$$

so

$$|\langle +z | +n \rangle|^2 + \tan^2 \frac{\theta}{2} |\langle +z | +n \rangle|^2 = 1 \quad (49)$$

$$|\langle +z | +n \rangle|^2 \left(1 + \tan^2 \frac{\theta}{2} \right) = 1 \quad (50)$$

$$|\langle +z | +n \rangle|^2 \left(\frac{1}{\cos^2 \frac{\theta}{2}} \right) = 1 \quad (51)$$

$$\langle +z | +n \rangle = \pm \cos \frac{\theta}{2} \quad (52)$$

We can plug this in to the equation for the two states (assuming +), the cosine will cancel, and we will get

$$\langle -z | +n \rangle = \sin \frac{\theta}{2} e^{i\phi} \quad (53)$$

Thus,

$$|+n\rangle = \cos \frac{\theta}{2} | +z \rangle + \sin \frac{\theta}{2} e^{i\phi} | -z \rangle$$

0.0.2 n= -1

$$\begin{pmatrix} \cos \theta + 1 & \sin \theta e^{-i\phi} \\ \sin \theta e^{i\phi} & -\cos \theta + 1 \end{pmatrix} \begin{pmatrix} \langle +z | -n \rangle \\ \langle -z | -n \rangle \end{pmatrix} = \vec{0} \quad (54)$$

$$(55)$$

The equation is

$$(\cos \theta + 1) \langle +z | -n \rangle = -\sin \theta e^{-i\phi} \langle -z | -n \rangle \quad (56)$$

$$(57)$$

We have similar trig identities

$$\cos \frac{\theta}{2} = \sqrt{\frac{1 + \cos \theta}{2}} \quad (58)$$

$$1 + \cos \theta = 2 \cos^2 \frac{\theta}{2} \quad (59)$$

$$\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \quad (60)$$

This gets

$$2 \cos^2 \frac{\theta}{2} \langle +z | -n \rangle = -2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} e^{i\phi} \langle -z | -n \rangle \quad (61)$$

We can now divide to get

$$\langle +z | -n \rangle = -2 \frac{\sin \frac{\theta}{2}}{\cos \frac{\theta}{2}} e^{-i\phi} \langle -z | -n \rangle \quad (62)$$

AGAIN (why is this problem so lonnnnggg TT)

$$|\langle +z | +n \rangle|^2 + |\langle -z | +n \rangle|^2 = 1 \quad (63)$$

So we have

$$|\langle -z | -n \rangle|^2 \left(1 + \tan^2 \frac{\theta}{2} \right) = 1 \quad (64)$$

$$|\langle -z | -n \rangle|^2 \left(\frac{1}{\cos^2 \frac{\theta}{2}} \right) = 1 \quad (65)$$

$$|\langle -z | -n \rangle|^2 = \cos^2 \frac{\theta}{2} \quad (66)$$

We can plug that in to the equation relating the amplitudes to get

$$\langle +z | -n \rangle = -\sin \frac{\theta}{2} e^{-i\phi} \quad (67)$$

The position of the overall phase doesn't matter, so I'll move it to the $-z$ position
Therefore, we have

$$|n\rangle = \sin \frac{\theta}{2} | +z \rangle + \cos \frac{\theta}{2} e^{i\phi} | -z \rangle$$

Townsend 3.5 (5 pts)

This problem demonstrates another way (also see Problem 3.2) to determine the eigenstates of $\hat{S}_n = \hat{\mathbf{S}} \cdot \mathbf{n}$. The operator

$$\hat{R}(\theta \mathbf{j}) = e^{-i\hat{S}_y\theta/\hbar}$$

rotates spin states by an angle θ counterclockwise about the y axis.

(a) Show that this rotation operator can be expressed in the form

$$\hat{R}(\theta \mathbf{j}) = \cos \frac{\theta}{2} - \frac{2i}{\hbar} \hat{S}_y \sin \frac{\theta}{2}$$

Suggestion: Use the states $|+\mathbf{z}\rangle$ and $|-\mathbf{z}\rangle$ as a basis. Express the operator $\hat{R}(\theta \mathbf{j})$ in matrix form by expanding $\hat{R}$ in a Taylor series. Examine the explicit form for the matrices representing $\hat{S}_y^2$, $\hat{S}_y^3$, and so on.

(b) Apply $\hat{R}$ in matrix form to the state $|+\mathbf{z}\rangle$ to obtain the state $|+\mathbf{n}\rangle$ given in Problem 3.2 with $\phi = 0$, that is, rotated by angle θ in the x - z plane. Show that $\hat{R}|-\mathbf{z}\rangle$ differs from $|-\mathbf{n}\rangle$ by an overall phase.

$$\hat{S}_y = \frac{\hbar}{2} \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (68)$$

$$\hat{S}_y^2 = \frac{\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad (69)$$

$$\hat{S}_y^3 = \hat{S}_y \quad (70)$$

$$(71)$$

We can Taylor expand:

$$e^{\frac{-i\hat{S}_y\theta}{\hbar}} = 1 - \frac{i^2 \hbar}{\hbar^2} \frac{\theta^2}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} + \frac{1}{2!} \left(\frac{i}{\hbar} \hat{S}_y \theta \right)^2 + \dots \quad (72)$$

Reduce the terms to either the identity or $\hat{S}_y$ for all powers of n . Take note of the terms $\frac{\hbar}{2}$ as a constant for each $\hat{S}_y$ matrix - this cancels the $\hbar$ from the prefactor terms.

$$\hat{R}(\theta \mathbf{j}) = - \underbrace{\frac{i}{\hbar} \hat{S}_y}_{\text{odd, sin}} \theta + \frac{1}{2!} \left(\frac{i}{\hbar} \frac{\hbar}{2} \frac{2}{\hbar} \hat{S}_y \theta \right)^2 + \dots \quad (73)$$

All the even power terms of $\hat{S}_y^{2n}$ are the identity, and all the odd terms are just $\hat{S}_y^{2n+1} = \hat{S}_y$. We can therefore separate out this expression to be

We can pull out the $\frac{i}{\hbar}$

Taking note of the Euler's identity that $e^{i\theta} = \cos \theta + i \sin \theta$. We have the Taylor series for $\cos \theta$. This becomes the Taylor expansion of

$$e^{-i\hat{S}_y \frac{\theta}{\hbar}} = \cos\left(\frac{\theta}{2}\right) \mathbf{I} + \frac{2i}{\hbar} \hat{S}_y \sin\left(\frac{\theta}{2}\right) \quad (74)$$

QED.

0.1 B

We can apply this matrix representation to $|+z\rangle$

$$R(\theta \mathbf{J}) = \begin{bmatrix} \cos \frac{\theta}{2} & 0 \\ 0 & \cos \frac{\theta}{2} \end{bmatrix} - \frac{2i}{\hbar} \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad (75)$$

$$= \begin{bmatrix} \cos \frac{\theta}{2} & \frac{2}{\hbar} \sin \frac{\theta}{2} \\ -\frac{2}{\hbar} \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{bmatrix} \quad (76)$$

applying this on $|+z\rangle$ by multiplying this matrix on the left side of

$$\hat{R}(\theta \mathbf{J}) \begin{pmatrix} |+z\rangle \\ |-z\rangle \end{pmatrix} \quad (77)$$

The top row is the $|+n\rangle$ vector, and the bottom row is the $|-n\rangle$ vector:

$$|+n\rangle = \cos \frac{\theta}{2} |+z\rangle + \frac{2}{\hbar} \sin \frac{\theta}{2} |-z\rangle \quad (78)$$

$$|-n\rangle = \frac{\hbar}{2} \sin \frac{\theta}{2} - \frac{2}{\hbar} \sin \frac{\theta}{2} |-z\rangle \quad (79)$$

We have gotten the result for $\phi = 0$

$$\begin{aligned} |+n\rangle &= \cos \frac{\theta}{2} |+z\rangle + e^{i\phi} \sin \frac{\theta}{2} |-z\rangle \\ |-n\rangle &= \sin \frac{\theta}{2} |+z\rangle - e^{i\phi} \cos \frac{\theta}{2} |-z\rangle \end{aligned}$$

Townsend 3.8 and 3.12 (5 pts)

Problem 3.8 Show that the operator $\hat{C}$ defined through $[\hat{A}, \hat{B}] = i\hat{C}$ is Hermitian, provided that the operators $\hat{A}$ and $\hat{B}$ are Hermitian. [This is fast.]

Problem 3.12 Verify that for a spin- $\frac{1}{2}$ particle that (a)

$$\hat{S}_z = \frac{\hbar}{2}|+\mathbf{z}\rangle\langle+\mathbf{z}| - \frac{\hbar}{2}|-\mathbf{z}\rangle\langle-\mathbf{z}|$$

and (b) the raising and lowering operators may be expressed as

$$\hat{S}_+ = \hbar|+\mathbf{z}\rangle\langle-\mathbf{z}| \quad \text{and} \quad \hat{S}_- = \hbar|-\mathbf{z}\rangle\langle+\mathbf{z}|$$

Note: It is sufficient to examine the action of these operators on the basis states $|+\mathbf{z}\rangle$ and $|-\mathbf{z}\rangle$, which of course form a complete set.

0.2 3.8

If a matrix $\hat{A}$ is Hermitian, then $\hat{A} = \hat{A}^\dagger$

We can expand out

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A} \quad (80)$$

$$= \hat{A}^\dagger \hat{B}^\dagger - \hat{B}^\dagger \hat{A}^\dagger \quad (81)$$

$$= [\hat{A}^\dagger, \hat{B}^\dagger] \quad (82)$$

Therefore,

$$i\hat{C} = i\hat{C}^\dagger \quad (83)$$

so $\hat{C}$ is Hermitian.

0.3 3.12

0.3.1 $\hat{S}_z$

$$\hat{S}_z|+\mathbf{z}\rangle = \frac{\hbar}{2}|+\mathbf{z}\rangle\langle+\mathbf{z}||+\mathbf{z}\rangle - 0 \quad (84)$$

$$= \frac{\hbar}{2}|+\mathbf{z}\rangle \quad (85)$$

$$\hat{S}_z|-\mathbf{z}\rangle = 0 - \frac{\hbar}{2}|-\mathbf{z}\rangle \quad (86)$$

$$= -\frac{\hbar}{2}|-\mathbf{z}\rangle \quad (87)$$

This validates the $\hat{S}_z$ expression.

We can check the raising and lowering operators now

0.3.2 $\hat{S}_+$

$$\hbar|+z\rangle\langle -z|$$

$$\hat{S}_+|-z\rangle = \hbar|+z\rangle \quad (88)$$

$$\hat{S}_+|+z\rangle = 0 \quad (89)$$

$$(90)$$

0.3.3 $\hat{S}_-$

$$\hbar|-z\rangle\langle +z|$$

$$\hat{S}_-|+z\rangle = \hbar|-z\rangle \quad (91)$$

$$\hat{S}_-|-z\rangle = 0 \quad (92)$$

Townsend 3.15 and 3.16 (5 pts)

(a) **Problem 3.15** Determine the eigenstates of $\hat{S}_x$ for a spin-1 particle in terms of the eigenstates $|1, 1\rangle$, $|1, 0\rangle$, and $|1, -1\rangle$ of $\hat{S}_z$.

(b) **Problem 3.16** A spin-1 particle exits an SGz device in a state with $S_z = \hbar$. The beam then enters an SGx device. What is the probability that the measurement of S_x yields the value 0?

0.4 A

We can write the $\hat{S}_x$ operator in the spin 1 $|+z\rangle$ basis

$$\hat{S}_x = \frac{\hbar}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad (93)$$

The eigenvalues are $\hbar, 0, -\hbar$. We can now solve for the vectors

0.4.1 $\lambda = \hbar$

$$\frac{1}{\sqrt{2}} \begin{bmatrix} -\sqrt{2} & 1 & 0 \\ 1 & -\sqrt{2} & 1 \\ 0 & 1 & -\sqrt{2} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0 \quad (94)$$

We get

$$-\sqrt{2}a + b = 0 \quad (95)$$

$$a - \sqrt{2}b + c = 0 \quad (96)$$

$$b - \sqrt{2}c = 0 \quad (97)$$

This has $a = c$ and $2a = \sqrt{2}b$
so the eigenvector is

$$\frac{1}{2} \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix} \quad (98)$$

0.4.2 $\lambda = 0$

The matrix is

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0 \quad (99)$$

This gives

$$b = 0 \quad (100)$$

$$a = -c \quad (101)$$

$$b = 0 \quad (102)$$

That gives

$$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad (103)$$

0.4.3 $-\hbar$

Finally,

$$\frac{1}{\sqrt{2}} \begin{bmatrix} \sqrt{2} & 1 & 0 \\ 1 & \sqrt{2} & 1 \\ 0 & 1 & \sqrt{2} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = 0 \quad (104)$$

This gives

$$\frac{1}{2} \begin{bmatrix} 1 \\ -\sqrt{2} \\ 1 \end{bmatrix} \quad (105)$$

0.5 B

(b) Problem 3.16 A spin-1 particle exits an SGz device in a state with $S_z = \hbar$. The beam then enters an SGx device. What is the probability that the measurement of S_x yields the value 0?

The particle is in the state $|+z\rangle$. This is asking

$$\langle S_x = 0 \text{ state} \mid 1, 1_z \rangle \quad (106)$$

We know that the eigenvector corresponding to $S_x|z\rangle = 0$ has the form

$$\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \quad (107)$$

from above. Therefore, we can write this as

$$\langle 1, 0_x \mid +z \rangle = \frac{1}{\sqrt{2}} [1, 0, -1] \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad (108)$$

$$= \frac{1}{\sqrt{2}} \quad (109)$$

Thus,

$$|\langle 1, 0_x \mid +z \rangle|^2 = \frac{1}{2} \quad (110)$$

Townsend 3.17 (5 pts)

A spin-1 particle is in the state

$$|\psi\rangle \xrightarrow{S_z \text{ basis}} \frac{1}{\sqrt{14}} \begin{pmatrix} 1 \\ 2 \\ 3i \end{pmatrix}$$

- (a) What are the probabilities that a measurement of S_z will yield the values $\hbar, 0$, or $-\hbar$ for this state? What is $\langle S_z \rangle$?
- (b) What is $\langle S_x \rangle$ for this state? *Suggestion:* Use matrix mechanics to evaluate the expectation value.
- (c) What is the probability that a measurement of S_x will yield the value $\hbar$ for this state?

0.6 a)

$$\begin{aligned} \hbar &: \frac{1}{14} \\ 0 &: \frac{2}{7} \\ -\hbar &: \frac{9}{14} \end{aligned}$$

$$\langle S_z \rangle = \frac{\hbar}{14} - \frac{9\hbar}{14} \quad (111)$$

$$= \frac{-8\hbar}{14} \quad (112)$$

$$= -\frac{4}{7}\hbar \quad (113)$$

0.7 b)We can write the $\hat{S}_x$ operator in the spin 1 $|+z\rangle$ basis

$$\hat{S}_x = \frac{\hbar}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad (114)$$

Where

$$|1, 1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad (115)$$

$$|1, 0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad (116)$$

$$|1, -1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (117)$$

We know that

$$\langle S_x \rangle = \langle \psi | S_x | \psi \rangle \quad (118)$$

Therefore, we can compute

$$\psi = \frac{1}{\sqrt{14}} \begin{pmatrix} 1 \\ 2 \\ 3i \end{pmatrix} \quad (119)$$

$$\psi' = \frac{1}{\sqrt{14}} (1, 2, -3i) \quad (120)$$

This just becomes

$$\langle S_x \rangle = \frac{1}{14} (1, 2, -3i) \frac{\hbar}{\sqrt{2}} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} 1 \\ 2 \\ 3i \end{pmatrix} \quad (121)$$

$$= \frac{1}{14} (1, 2, -3i) \frac{\hbar}{\sqrt{2}} \begin{bmatrix} 2 \\ 1+3i \\ 2 \end{bmatrix} \quad (122)$$

$$= \frac{\hbar}{14\sqrt{2}} (2 + 2 + 6i - 6i) \quad (123)$$

$$= \frac{4\hbar}{14\sqrt{2}} \quad (124)$$

0.8 C

The state for $\hat{S}_x = \hbar$ has the matrix representation

$$|\hat{S}_x = \hbar\rangle = \frac{1}{2} \begin{bmatrix} 1 \\ \sqrt{2} \\ 1 \end{bmatrix} \quad (125)$$

We can write this now as

$$\langle +x | \psi \rangle = \frac{1}{2} [1, \sqrt{2}, -1] \frac{1}{\sqrt{14}} \begin{pmatrix} 1 \\ 2 \\ 3i \end{pmatrix} \quad (126)$$

$$= \frac{1}{2\sqrt{14}} (1 + 2\sqrt{2} - 3i) \quad (127)$$

The probability is this squared, so

$$|\langle +x | \psi \rangle|^2 = \frac{1}{56} ((1 + 2\sqrt{2})^2 + 9) \quad (128)$$

$$= \frac{1}{56} (18 + 4\sqrt{2}) \quad (129)$$

This feels like a weird number, but okay??? I can do algebra I think?

Townsend 3.20 (5 pts)

A beam of spin-1 particles is sent through a series of three Stern-Gerlach measuring devices (Fig. 3.11). The first SGz device transmits particles with $S_z = \hbar$ and filters out particles with $S_z = 0$ and $S_z = -\hbar$. The second device, an SGn device, transmits particles with $S_n = \hbar$ and filters out particles with $S_n = 0$ and $S_n = -\hbar$, where the axis **n** makes an angle θ in the x - z plane with respect to the z axis. A last SGz device transmits particles with $S_z = -\hbar$ and filters out particles with $S_z = \hbar$ and $S_z = 0$.

- What fraction of the particles transmitted by the first SGz device will survive the third measurement? *Note:* The states with $S_n = \hbar$, $S_n = 0$, and $S_n = -\hbar$ in the S_z basis follow directly from applying the rotation operator given in Problem 3.19 to states with $S_z = \hbar$, $S_z = 0$, and $S_z = -\hbar$, respectively.
- How must the angle θ of the SGn device be oriented so as to maximize the number of particles that are transmitted by the final SGz device? What fraction of the particles survive the third measurement for this value of θ ?
- What fraction of the particles survive the last measurement if the SGn device is removed from the experiment?

Repeat your calculation for parts (a), (b), and (c) if the last SGz device transmits particles with $S_z = 0$ only.

0.8.1 A

The rotation matrix is given by

$$\hat{R}(\theta) = \begin{pmatrix} \frac{1+\cos\theta}{2} & -\frac{\sin\theta}{\sqrt{2}} & \frac{1-\cos\theta}{2} \\ \frac{\sin\theta}{\sqrt{2}} & \cos\theta & -\frac{\sin\theta}{\sqrt{2}} \\ \frac{1-\cos\theta}{2} & \frac{\sin\theta}{\sqrt{2}} & \frac{1+\cos\theta}{2} \end{pmatrix} \quad (130)$$

This tells us that

$$|n\rangle = \hat{R}(\theta)|z\rangle \quad (131)$$

For $|S_n = \hbar\rangle$, we multiply by the basis vector

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad (132)$$

This gets

$$|+n\rangle = \begin{pmatrix} \frac{1+\cos\theta}{2} \\ \frac{\sin\theta}{\sqrt{2}} \\ \frac{1-\cos\theta}{2} \end{pmatrix} \quad (133)$$

We can now find

$$|\langle +n | +z \rangle|^2 = \left[\frac{1+\cos\theta}{2}, \frac{\sin\theta}{\sqrt{2}}, \frac{1-\cos\theta}{2} \right] \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad (134)$$

$$= \left(\frac{1+\cos\theta}{2} \right)^2 \quad (135)$$

We have the next step as

$$|\langle -z | +n \rangle|^2 = \left([0, 0, 1] \begin{pmatrix} \frac{1+\cos\theta}{2} \\ \frac{\sin\theta}{\sqrt{2}} \\ \frac{1-\cos\theta}{2} \end{pmatrix} \right)^2 \quad (136)$$

$$= \left(\frac{1-\cos\theta}{2} \right)^2 \quad (137)$$

So the odds of going through both are

$$\left(\frac{1 - \cos \theta}{2}\right)^2 \left(\frac{1 + \cos \theta}{2}\right)^2 \quad (138)$$

$$= \left(\frac{1 + \cos^2 \theta}{4}\right)^2 \quad (139)$$

0.8.2 B

at $\theta = \pi$. The arguments are maximized by $\cos^2 \theta$

0.8.3 C

None would pass through, since $|S_z = \hbar\rangle$ has no transition and can't be immediately in the state $|S_z = -\hbar\rangle$ or $|S_z = 0\rangle$.

0.8.4 D?

A We would now have
 $|S_z = 0, +n\rangle$

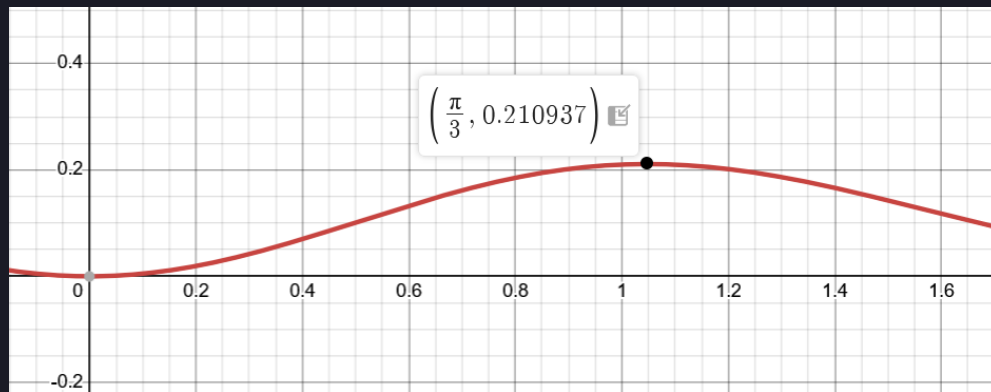
$$|+n\rangle = \begin{pmatrix} \frac{1+\cos \theta}{2} \\ \frac{\sin \theta}{\sqrt{2}} \\ \frac{1-\cos \theta}{2} \end{pmatrix} \quad (140)$$

So this would be $\left(\frac{\sin \theta}{\sqrt{2}}\right)^2$

this gets

$$\left(\frac{1 + \cos \theta}{2}\right)^2 \cdot \left(\frac{\sin \theta}{\sqrt{2}}\right)^2 \quad (141)$$

B this is maximized at
 $\theta = \frac{\pi}{3}$



C This would stay the same - we don't have any $S_z = 0$ if we have just enforced $S_z = \hbar$.

1 Extra work that I didn't want to delete because I was proud but it was useless

1.1 a

We are computing here

$$|\langle S_n = \hbar \mid S_z = \hbar \rangle|^2 \cdot |\langle S_z = -\hbar \mid S_n = \hbar \rangle|^2 \quad (142)$$

$$|1, 1\rangle_z = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\langle 1 \mid 1 \rangle_n = R(\theta) |1, 1\rangle_z \quad (143)$$

$$(144)$$

$$J_y = \frac{1}{2i}(J_+ - J_-)$$

Lets use the basis

$$|1, +1\rangle \rightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \quad (145)$$

$$|1, 0\rangle \rightarrow \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \quad (146)$$

$$|1, -1\rangle \rightarrow \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad (147)$$

$$J_+ |j, m\rangle = \hbar \sqrt{j(j+1) - m(m+1)} |j, m+1\rangle$$

$$J_+ |1, 1\rangle = 0 \quad (148)$$

$$J_+ |1, 0\rangle = \hbar \sqrt{2} |1, 1\rangle \quad (149)$$

$$J_+ |1, -1\rangle = \hbar \sqrt{2} |1, 0\rangle \quad (150)$$

So we can make the J_+ matrix:

$$J_+ = \hbar \begin{bmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{bmatrix} \quad (151)$$

$$\hat{J}_- = \hbar \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix} \quad (152)$$

$$\hat{J}_y = \frac{1}{2i} \begin{bmatrix} 0 & \sqrt{2} & 0 \\ -\sqrt{2} & 0 & \sqrt{2} \\ 0 & -\sqrt{2} & 0 \end{bmatrix} \quad (153)$$

This gives us the generator of rotations, that

$$|+n\rangle = e^{\frac{-i}{\hbar} \hat{J}_y \theta} |+\rangle \quad (154)$$

The pavlovian urge is to ~~~integrate~~~ taylor expand

$$e^{\frac{-i}{\hbar} \hat{J}_y \theta} = 1 - \quad (155)$$

2 a different question

We can write the $\hat{S}_x$ operator in the spin 1 $|+z\rangle$ basis

$$\hat{S}_x = \frac{\hbar}{2} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \quad (156)$$

This gives us that

$$\frac{\hbar}{2} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad (157)$$

$$\frac{\hbar}{2} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \quad (158)$$

$$\frac{\hbar}{2} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad (159)$$

$$(160)$$

Summarized,

$$\hat{S}_x |1, 1\rangle = \frac{\hbar}{2} |1, 0\rangle \quad (161)$$

$$\hat{S}_x |1, 0\rangle = \frac{\hbar}{2} (|1, 1\rangle + |1, -1\rangle) \quad (162)$$

$$\hat{S}_x |1, -1\rangle = \frac{\hbar}{2} |1, 0\rangle \quad (163)$$

$$(164)$$